MSG No. 177.154 P_c622 [Type IV, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `622`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `62'2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`

Table 3: Wyckoff site: `4c`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,3,5,10,11,12]`
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[2,4,6,7,8,9]`
3	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[13,15,17,22,23,24]`
4	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[14,16,18,19,20,21]`

Table 4: Wyckoff site: `4d`, site symmetry: `3.2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,3,5,22,23,24]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[2,4,6,19,20,21]`
3	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[7,8,9,14,16,18]`
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[10,11,12,13,15,17]`

Table 5: Wyckoff site: `4e`, site symmetry: `6..`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,4,5,6]`
2	$[0, 0, -z]$	`[7,8,9,10,11,12]`
3	$\left[0, 0, z + \frac{1}{2}\right]$	`[13,14,15,16,17,18]`
4	$\left[0, 0, \frac{1}{2} - z\right]$	`[19,20,21,22,23,24]`

Table 6: Wyckoff site: `6f`, site symmetry: `222`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,7,11]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,5,9,10]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,6,8,12]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[13,16,19,23]
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[14,17,21,22]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[15,18,20,24]

Table 7: Wyckoff site: 6g, site symmetry: 22'2'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[1,4,19,23]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[2,5,21,22]
3	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[3,6,20,24]
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[7,11,13,16]
5	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[8,12,15,18]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[9,10,14,17]

Table 8: Wyckoff site: 8h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,3,5]
2	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[2,4,6]
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[7,8,9]
4	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[10,11,12]
5	$\left[\frac{1}{3}, \frac{2}{3}, z+\frac{1}{2}\right]$	[13,15,17]
6	$\left[\frac{2}{3}, \frac{1}{3}, z+\frac{1}{2}\right]$	[14,16,18]
7	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}-z\right]$	[19,20,21]
8	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}-z\right]$	[22,23,24]

Table 9: Wyckoff site: 12i, site symmetry: 2..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,4]
2	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[2,5]
3	$\left[0, \frac{1}{2}, z\right]$	[3,6]
4	$\left[\frac{1}{2}, 0, -z\right]$	[7,11]
5	$\left[0, \frac{1}{2}, -z\right]$	[8,12]
6	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[9,10]
7	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[13,16]
8	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	[14,17]
9	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[15,18]
10	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[19,23]
11	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[20,24]
12	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	[21,22]

Table 10: Wyckoff site: `12j`, site symmetry: `.2.`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,7]`
2	$[x,\ x,\ 0]$	`[2,10]`
3	$[0,\ x,\ 0]$	`[3,8]`
4	$[-x,\ 0,\ 0]$	`[4,11]`
5	$[-x,\ -x,\ 0]$	`[5,9]`
6	$[0,\ -x,\ 0]$	`[6,12]`
7	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[13,19]`
8	$\left[x,\ x,\ \frac{1}{2}\right]$	`[14,22]`
9	$\left[0,\ x,\ \frac{1}{2}\right]$	`[15,20]`
10	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[16,23]`
11	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[17,21]`
12	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[18,24]`

Table 11: Wyckoff site: `12k`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,19]`
2	$\left[x,\ x,\ \frac{1}{4}\right]$	`[2,22]`
3	$\left[0,\ x,\ \frac{1}{4}\right]$	`[3,20]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,23]`
5	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[5,21]`
6	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[6,24]`
7	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[7,13]`
8	$\left[0,\ x,\ \frac{3}{4}\right]$	`[8,15]`
9	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[9,17]`
10	$\left[x,\ x,\ \frac{3}{4}\right]$	`[10,14]`
11	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[11,16]`
12	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[12,18]`

Table 12: Wyckoff site: `12l`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ -x,\ 0]$	`[1,12]`
2	$[2x,\ x,\ 0]$	`[2,7]`
3	$[x,\ 2x,\ 0]$	`[3,10]`
4	$[-x,\ x,\ 0]$	`[4,8]`
5	$[-2x,\ -x,\ 0]$	`[5,11]`
6	$[-x,\ -2x,\ 0]$	`[6,9]`
7	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[13,24]`
8	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[14,19]`
9	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[15,22]`
10	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[16,20]`
11	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[17,23]`
12	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[18,21]`

Table 13: Wyckoff site: `12m`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\, -x,\, \frac{1}{4}\right]$	`[1,24]`
2	$\left[2x,\, x,\, \frac{1}{4}\right]$	`[2,19]`
3	$\left[x,\, 2x,\, \frac{1}{4}\right]$	`[3,22]`
4	$\left[-x,\, x,\, \frac{1}{4}\right]$	`[4,20]`
5	$\left[-2x,\, -x,\, \frac{1}{4}\right]$	`[5,23]`
6	$\left[-x,\, -2x,\, \frac{1}{4}\right]$	`[6,21]`
7	$\left[2x,\, x,\, \frac{3}{4}\right]$	`[7,14]`
8	$\left[-x,\, x,\, \frac{3}{4}\right]$	`[8,16]`
9	$\left[-x,\, -2x,\, \frac{3}{4}\right]$	`[9,18]`
10	$\left[x,\, 2x,\, \frac{3}{4}\right]$	`[10,15]`
11	$\left[-2x,\, -x,\, \frac{3}{4}\right]$	`[11,17]`
12	$\left[x,\, -x,\, \frac{3}{4}\right]$	`[12,13]`

Table 14: Wyckoff site: `24n`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[x-y,\, x,\, z]$	`[2]`
3	$[-y,\, x-y,\, z]$	`[3]`
4	$[-x,\, -y,\, z]$	`[4]`
5	$[-x+y,\, -x,\, z]$	`[5]`
6	$[y,\, -x+y,\, z]$	`[6]`
7	$[x-y,\, -y,\, -z]$	`[7]`
8	$[y,\, x,\, -z]$	`[8]`
9	$[-x,\, -x+y,\, -z]$	`[9]`
10	$[x,\, x-y,\, -z]$	`[10]`
11	$[-x+y,\, y,\, -z]$	`[11]`
12	$[-y,\, -x,\, -z]$	`[12]`
13	$\left[x,\, y,\, z+\frac{1}{2}\right]$	`[13]`
14	$\left[x-y,\, x,\, z+\frac{1}{2}\right]$	`[14]`
15	$\left[-y,\, x-y,\, z+\frac{1}{2}\right]$	`[15]`
16	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	`[16]`
17	$\left[-x+y,\, -x,\, z+\frac{1}{2}\right]$	`[17]`
18	$\left[y,\, -x+y,\, z+\frac{1}{2}\right]$	`[18]`
19	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	`[19]`
20	$\left[y,\, x,\, \frac{1}{2}-z\right]$	`[20]`
21	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	`[21]`
22	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	`[22]`
23	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	`[23]`
24	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	`[24]`